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SUMMER SCHOOL - INTRODUCTION TO COMPLEX SYSTEMS

Feedbacks & Emergence

Slides available at XXX

Dr. Robbin Bastiaansen
r.bastiaansen@uu.nl





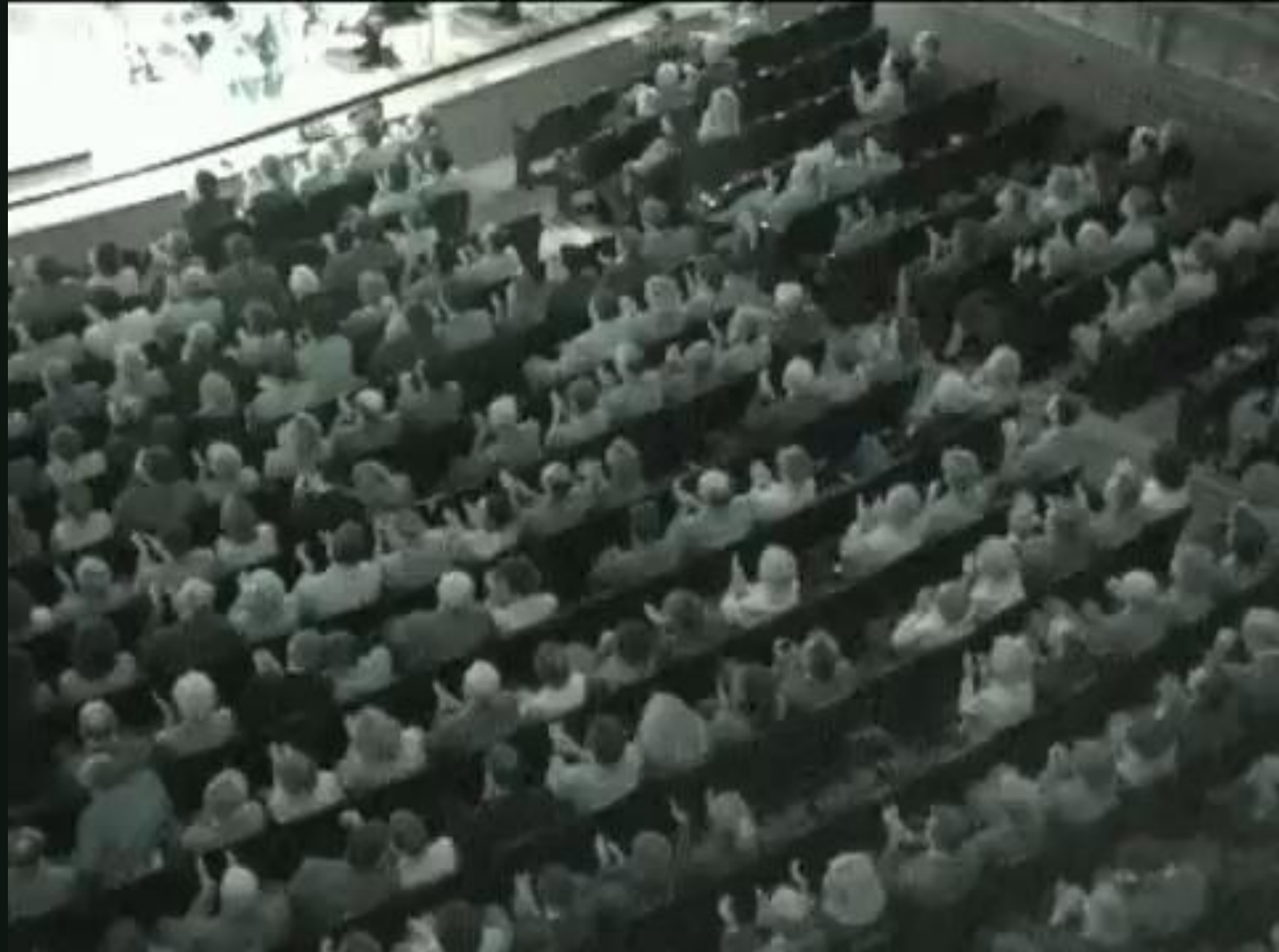
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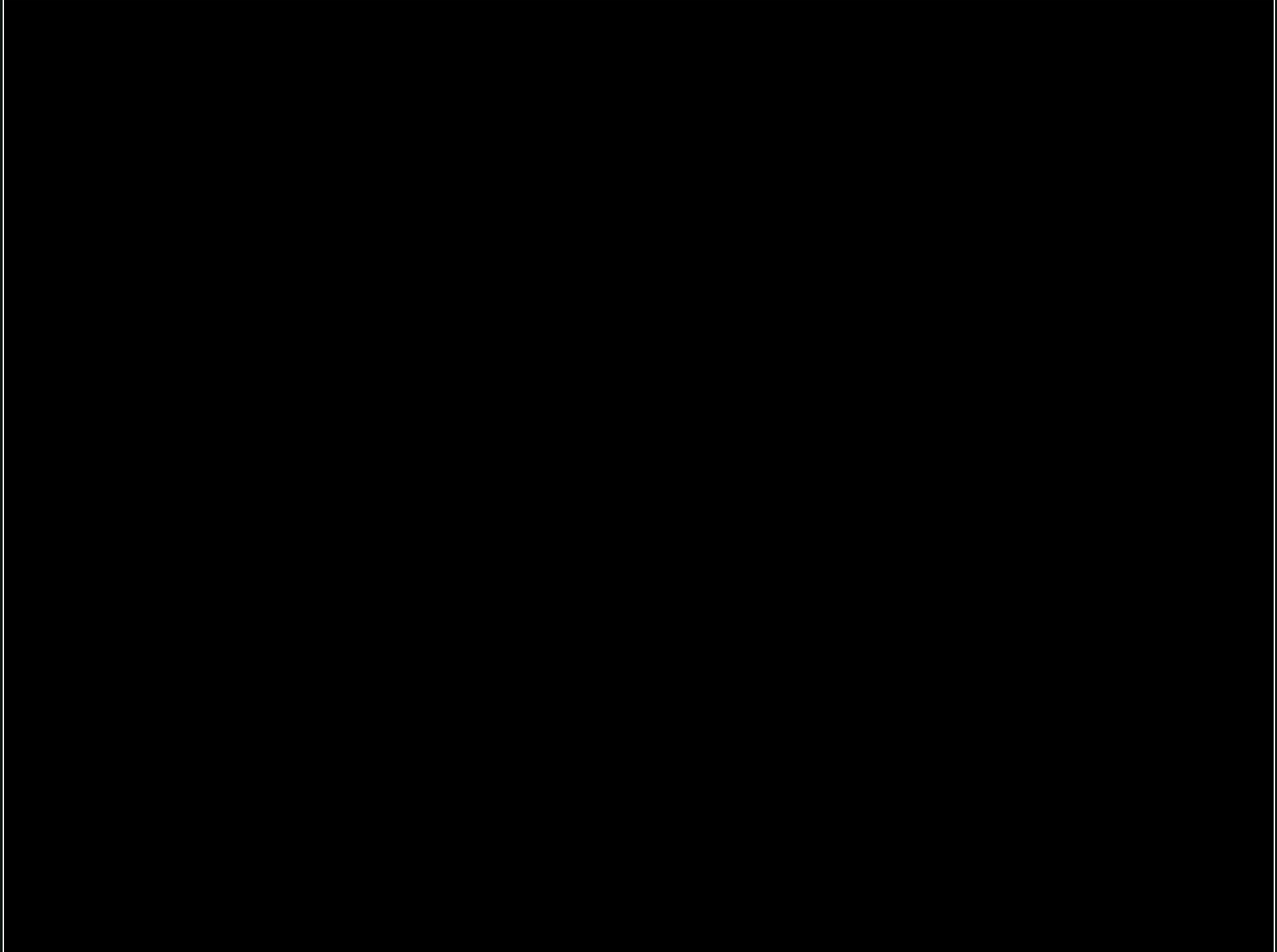
A close-up photograph of a large number of ants, likely red ants, crawling along a green leaf. The ants are densely packed in some areas, forming a long line that stretches across the frame. The background is a soft, out-of-focus green, suggesting a natural outdoor setting.

Part I Examples & Definition









Some definitions

A **complex system** is a (typically large) collection of (typically similar) elements that are connected to each other by 'signals'.

'signals' could be e.g. money, energy, agreements, hormones, information, tweets, infection, friendship

Emergence

Individuals perceive signals from other individuals. Then, a sufficiently summed strength of these signals causes individuals to act in certain ways that produce additional signal, leading to emergent collective behaviour.

Crucially, NO central control. It is self-organised!



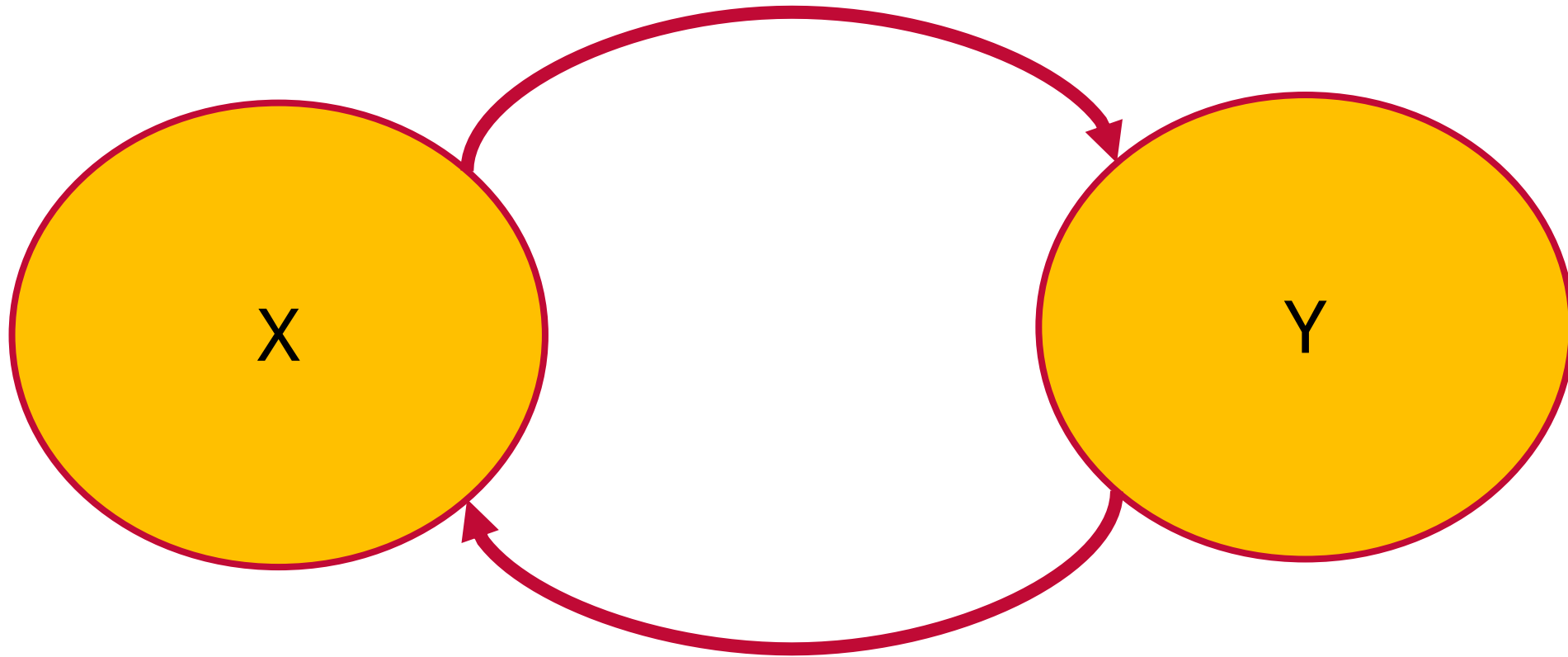
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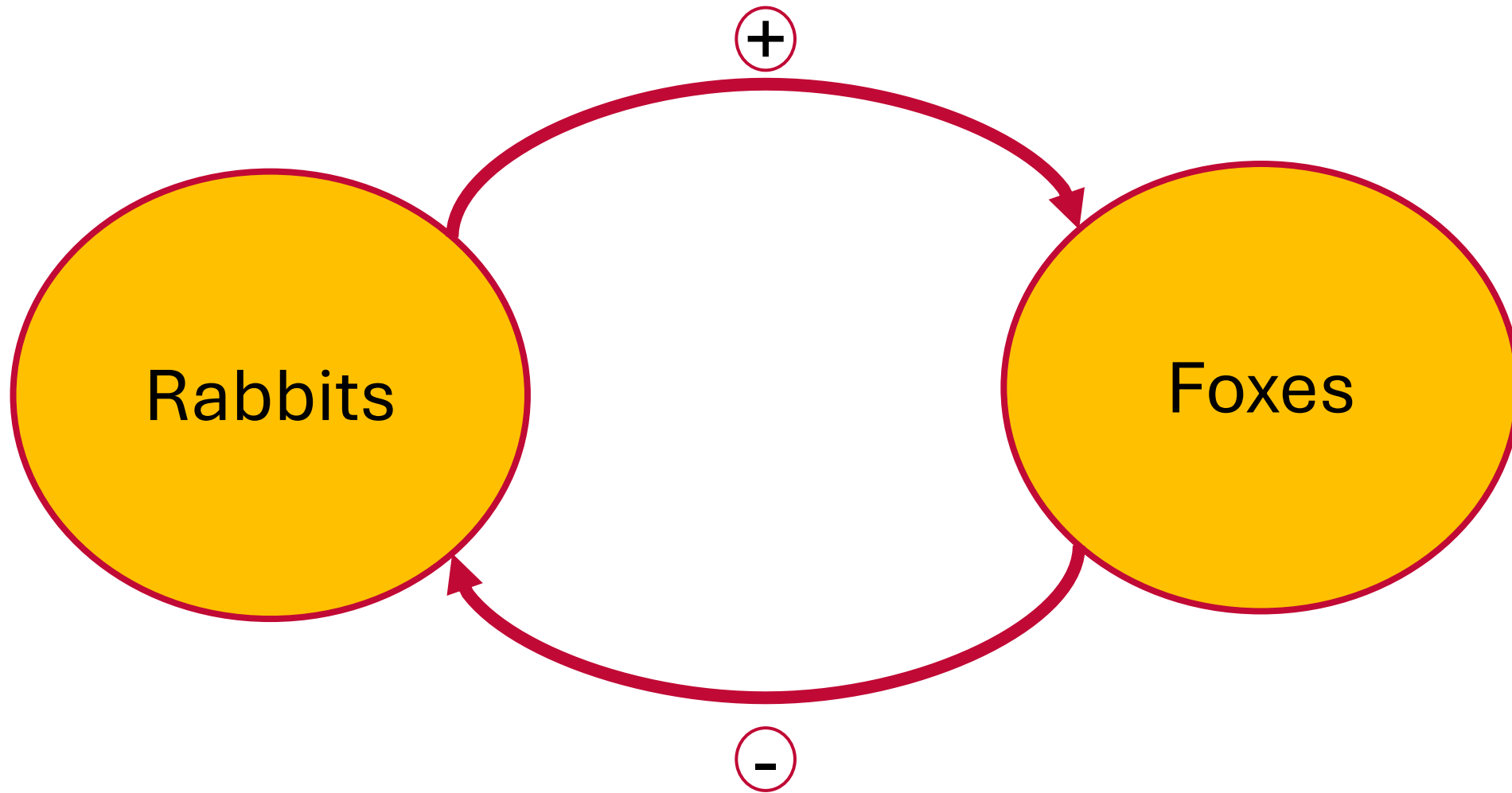
A close-up photograph of a large number of ants, likely Lasius niger, moving across a green leaf. The ants are densely packed in some areas, forming a line, and more spread out in others. Their bodies are a reddish-brown color, and their legs are thin and jointed. The leaf is a vibrant green with visible veins.

Part II *Feedbacks*

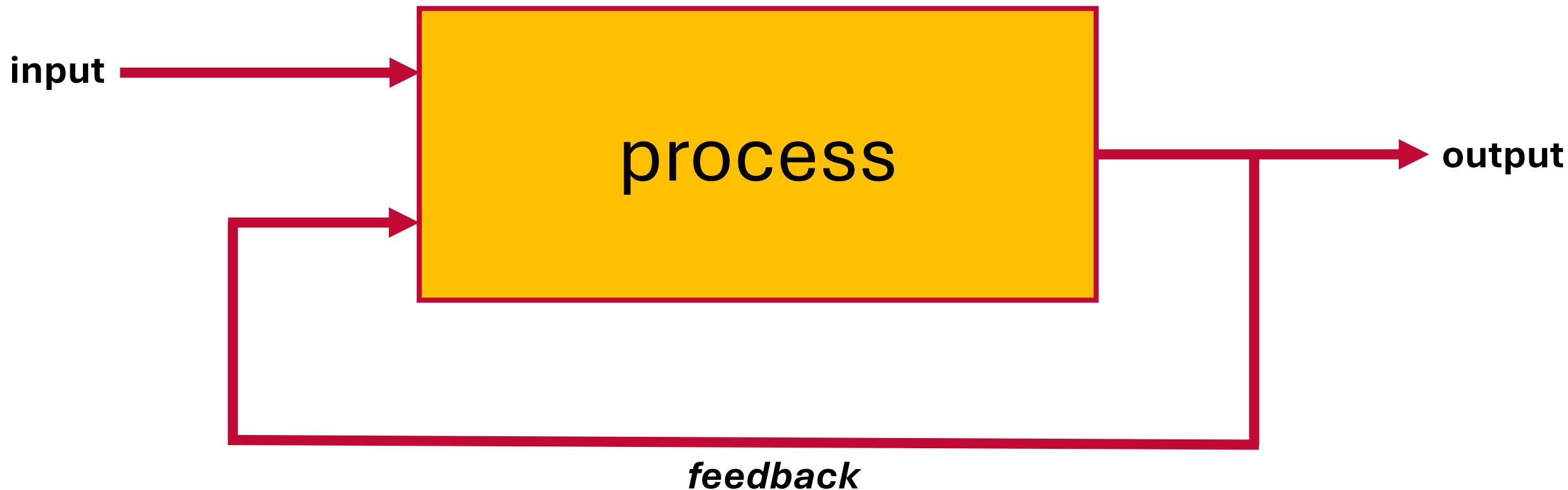
Feedback loops



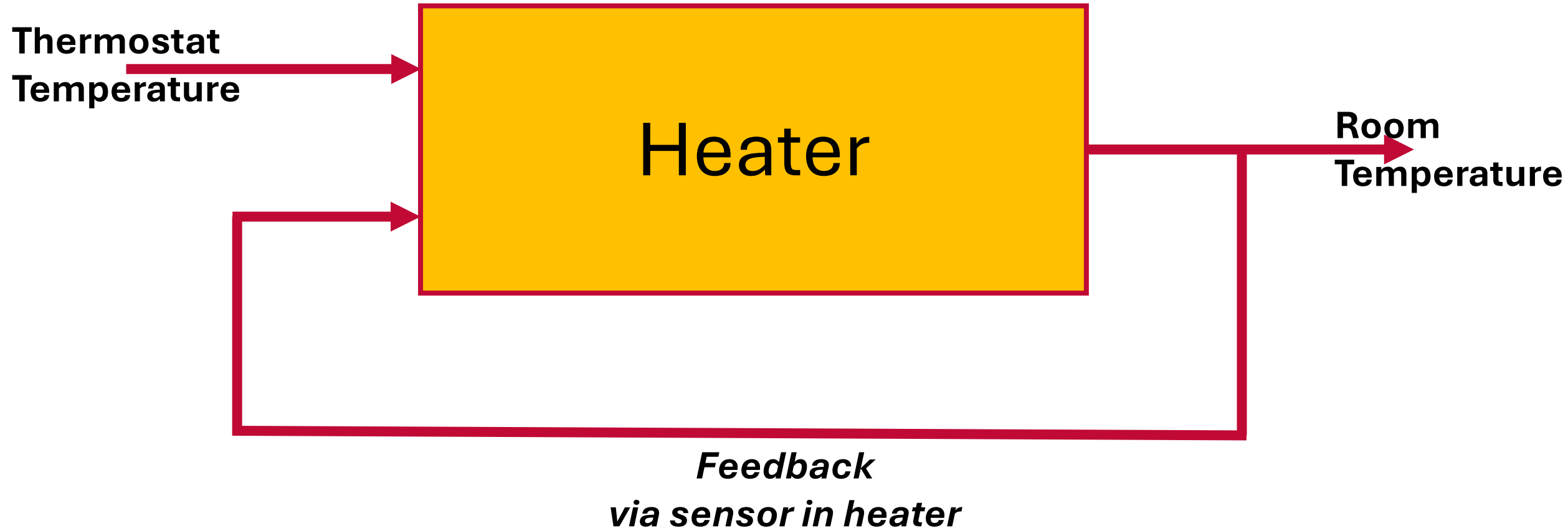
Feedback loops – Example predator-prey system



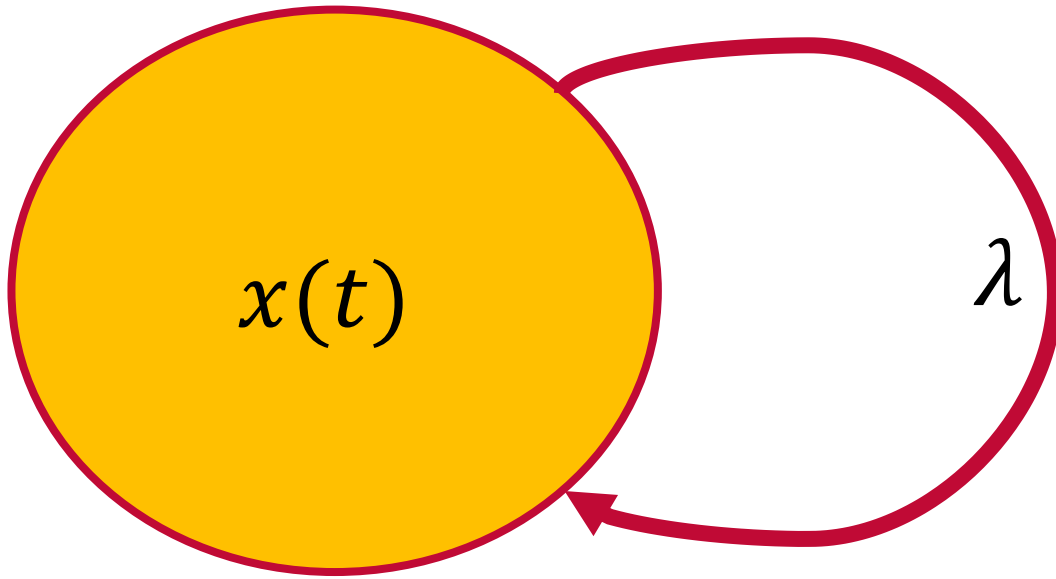
Feedback loops – classic control theory view



Feedback loops – classic – example



Dynamics



Generalises with multiple components:

$$\frac{d\vec{x}}{dt} = A \vec{x}$$

and eigenvalues of matrix A

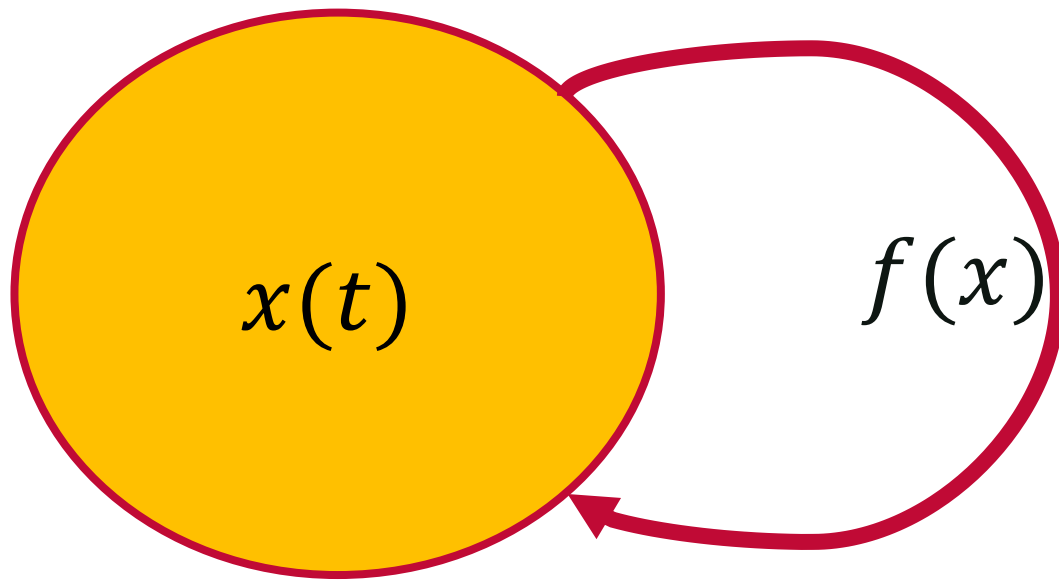
$$\frac{dx}{dt}(t) = \lambda x(t)$$

$$x(t) = e^{\lambda t} x_0$$

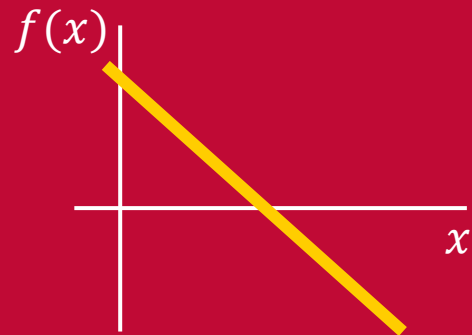
$Re \lambda < 0 : x(t) \rightarrow 0$
negative feedback

$Re \lambda > 0 : x(t) \rightarrow \infty$
positive feedback

Nonlinear feedbacks



Example:
 $f(x) = 1 - x$



$$\frac{dx}{dt}(t) = f(x)x =: g(x)$$



Negative feedback: stabilises
Positive feedback: destabilises



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A close-up photograph of a large number of ants, likely red ants, crawling along a green leaf. The ants are densely packed in some areas, forming a line that follows the edge of the leaf. The background is a soft, out-of-focus green.

Part III *Mean Field* *Approximations*

Connecting micro to macro



MODEL

state $s_j(t)$

0 : no vegetation

1 : vegetation

Windblown

$$\mathbb{P}(0 \rightarrow 1) = \epsilon$$

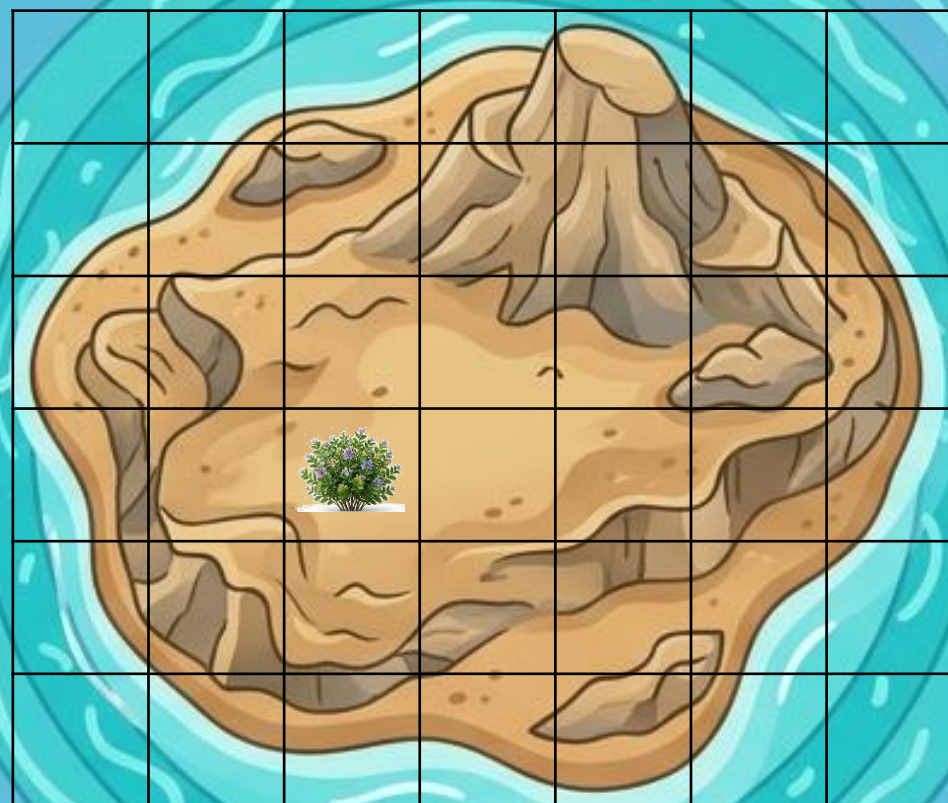
Reproduction

$$\mathbb{P}(0 \rightarrow 1) = \alpha \rho^2$$

with ρ density around site

Mortality

$$\mathbb{P}(1 \rightarrow 0) = \delta$$



Mean-field approximation



MODEL

state $s_j(t)$

0 : no vegetation

1 : vegetation

Windblown

$$\mathbb{P}(0 \rightarrow 1) = \epsilon$$

Reproduction

$$\mathbb{P}(0 \rightarrow 1) = \alpha \rho^2$$

with ρ density around site

Mortality

$$\mathbb{P}(1 \rightarrow 0) = \delta$$

Mean field approximation

Let $S(t) := \frac{1}{N} \sum_{j=1}^N s_j(t)$ and let $x(t) := \mathbb{E}[S(t)]$

Then:

$$\begin{aligned} x(t+1) &= \mathbb{E}[S(t+1)] \\ &= \mathbb{E}[s(t) = 1] \cdot \mathbb{P}(1 \rightarrow 1) + \mathbb{E}[s(t) = 0] \cdot \mathbb{P}(0 \rightarrow 1) \\ &= x(t) \cdot (1 - \delta) + (1 - x(t)) \cdot (\epsilon + \alpha \mathbb{E}[\rho^2]) \\ &= x(t) \cdot (1 - \delta) + (1 - x(t)) \cdot (\epsilon + \alpha x^2) \end{aligned}$$

In which we assume spatial mixing, i.e. $\mathbb{E}[\rho^2] = x^2$

$$\text{So: } \frac{dx}{dt} = \epsilon - (\epsilon + \delta)x + \alpha x^2 - \alpha x^3$$

Analysis of mean-field equation



$$\frac{dx}{dt} = \epsilon - (\epsilon + \delta)x + \alpha x^2 - \alpha x^3$$

Using dimensional analysis this reduces to

$$\frac{dy}{dt} = v - y + y^2 - r y^3$$

with

$$v := \frac{\epsilon\alpha}{(\epsilon + \delta)^2}, r := \frac{\epsilon + \delta}{\alpha}, y := \frac{x}{r}$$

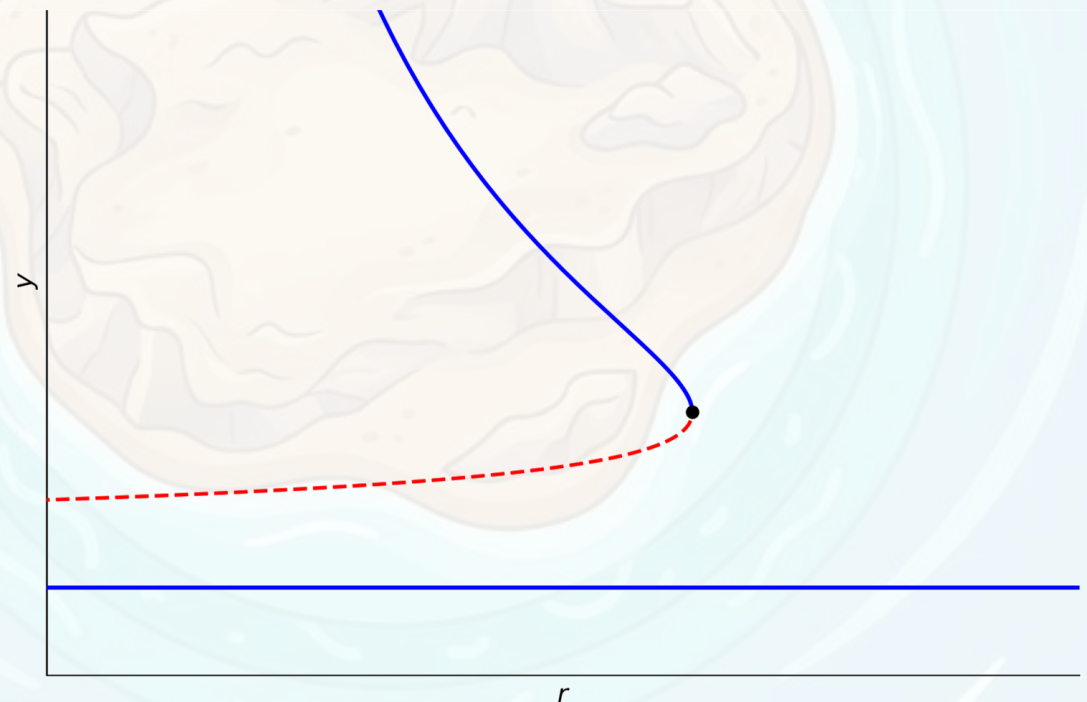
We take $v = 0$ ($\epsilon = 0$) and study

$$\frac{dy}{dt} = -y + y^2 - r y^3$$

$$\frac{dy}{dt} = -y + y^2 - r y^3$$

Fixed points:

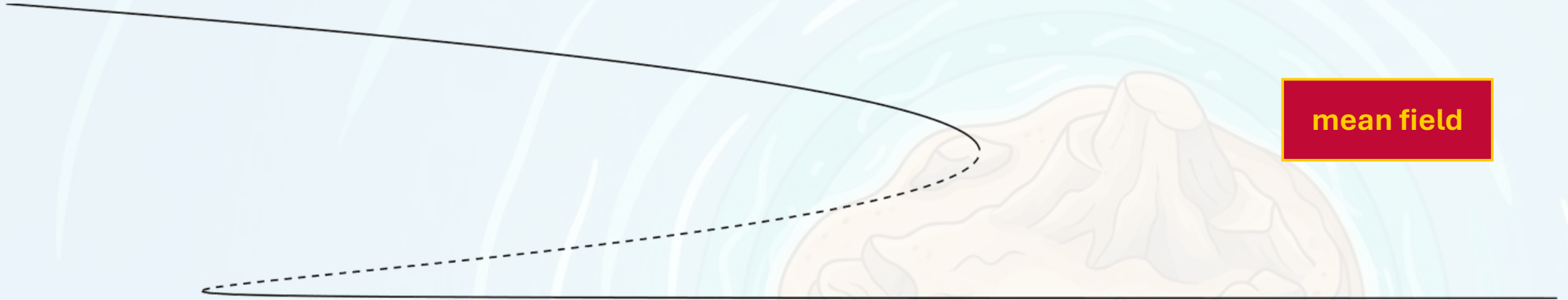
$$y_1 = 0, y_{2,3} = \frac{1 \pm \sqrt{1-4r}}{2r}$$



Compare to simulation

Parameters:

$$\epsilon = 0.001, \alpha := 1.50$$



Other bifurcations

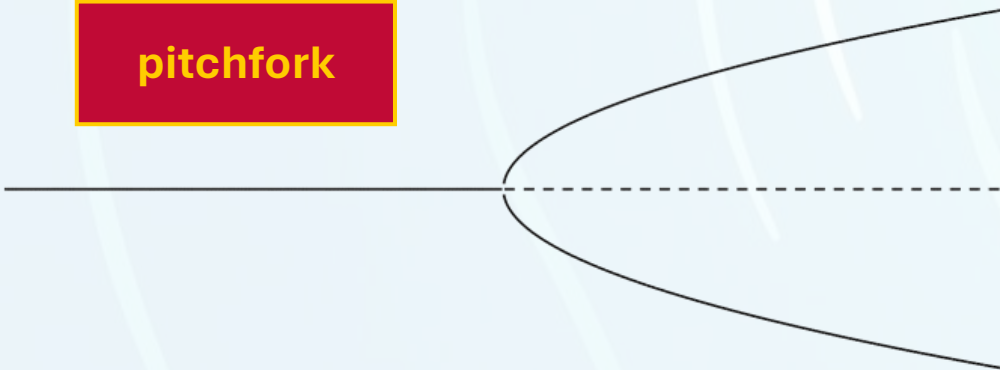
fold



transcritical

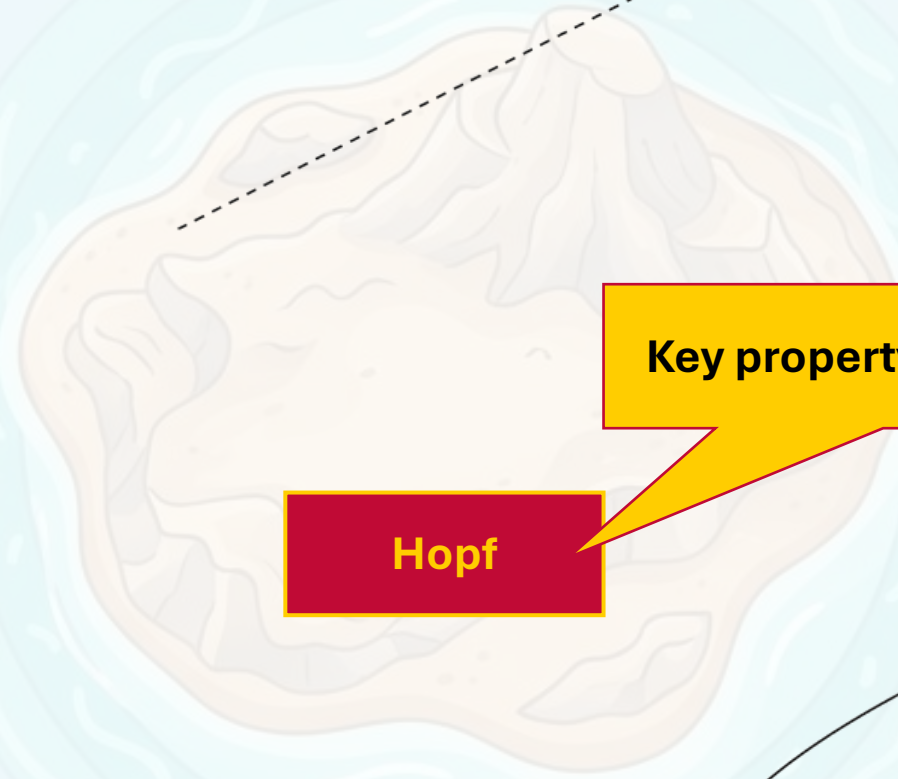


pitchfork



Hopf

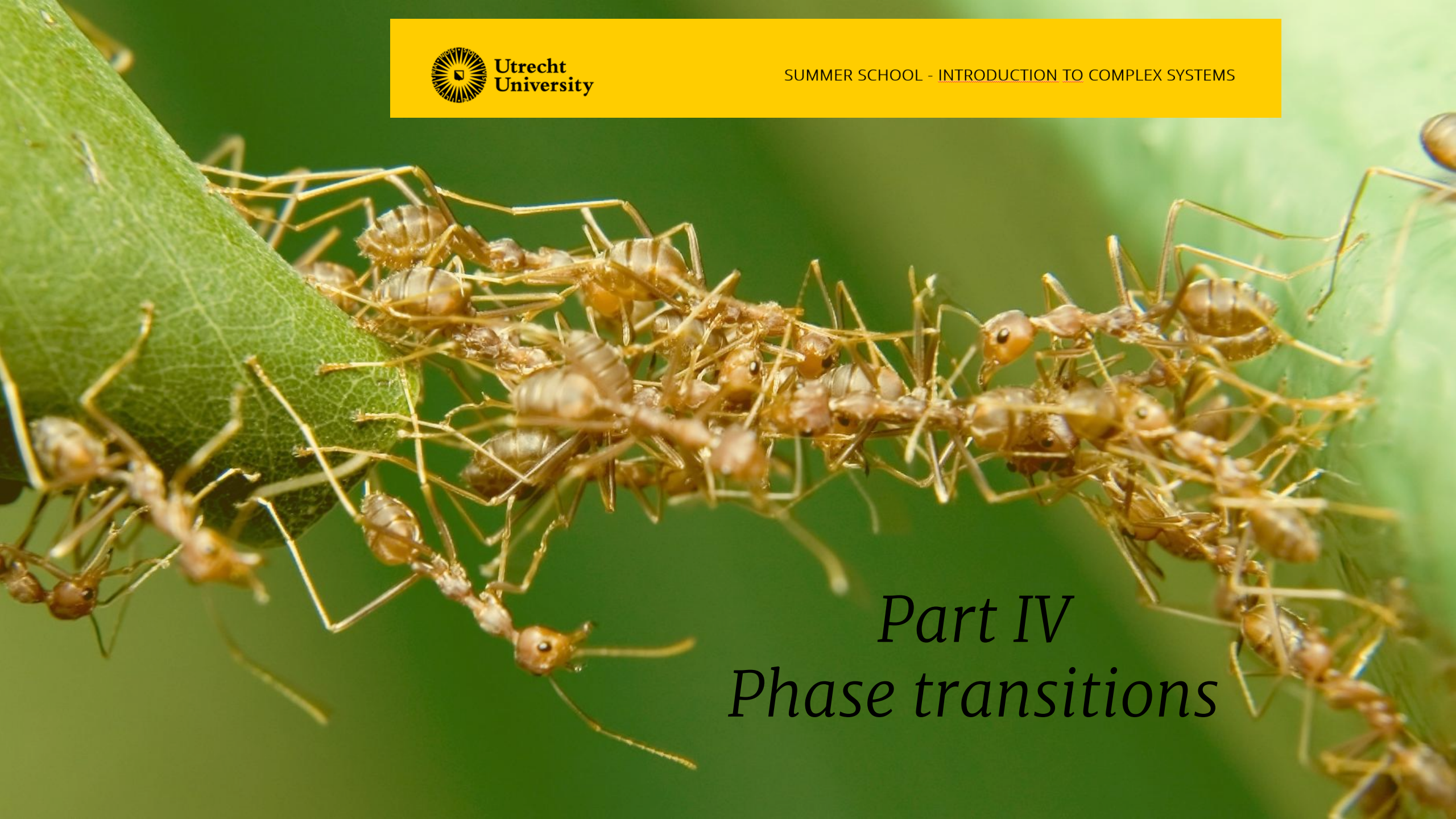
Key property is oscillations





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Part IV

Phase transitions

Simplified Ising Model

MODEL

state $s_j(t)$

1 : spin up

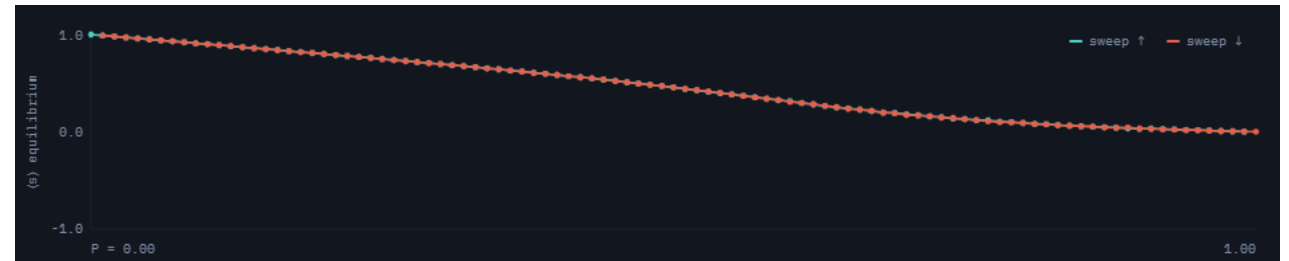
-1 : spin down

Update rule

- with probability P : flip cell
- Otherwise:
 - count spin of neighbours
 - add bias H
 - If sum is ≥ 0 : cell will be +1
 - If sum is < 0 : cell will be -1

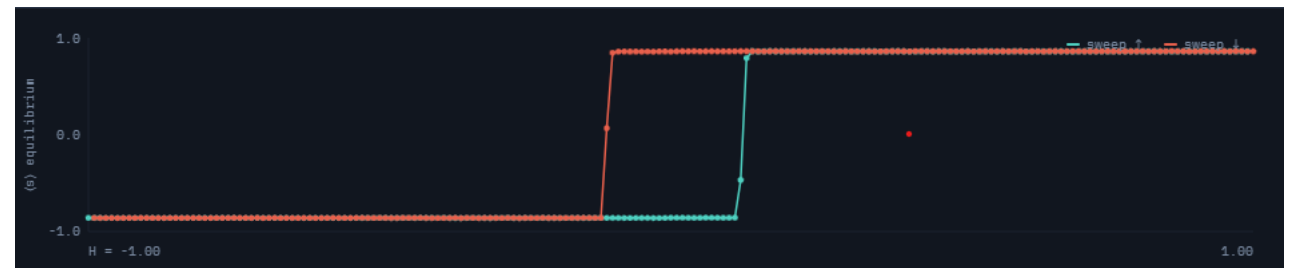
Varying P
($H = 0$)

second-order phase
transition



Varying H
($P = 0, 14$)

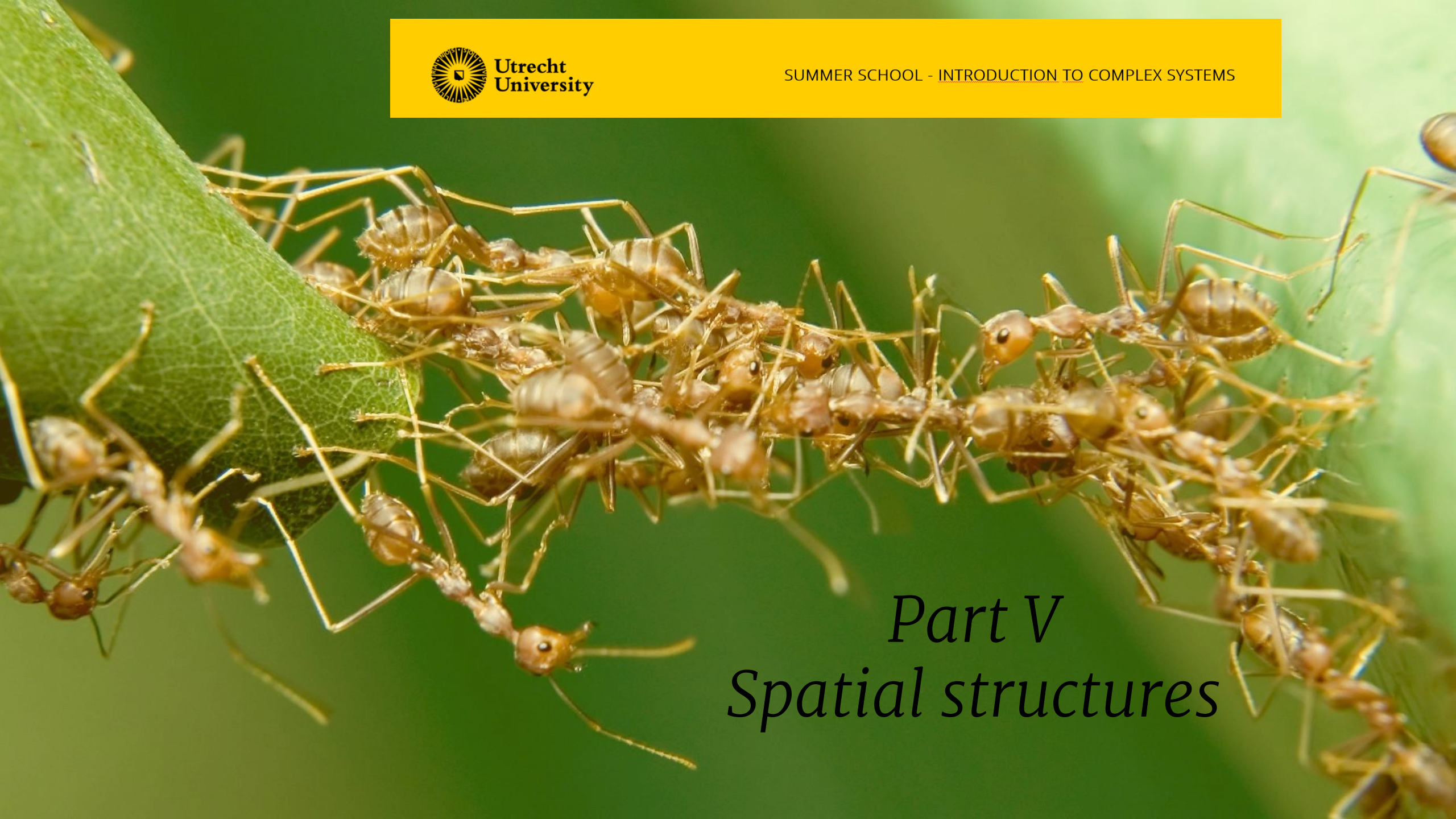
first-order phase
transition





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Part V *Spatial structures*

Conway's Game of Life

MODEL

state $s_j(t)$

1 : ON

0 : OFF

Update rule

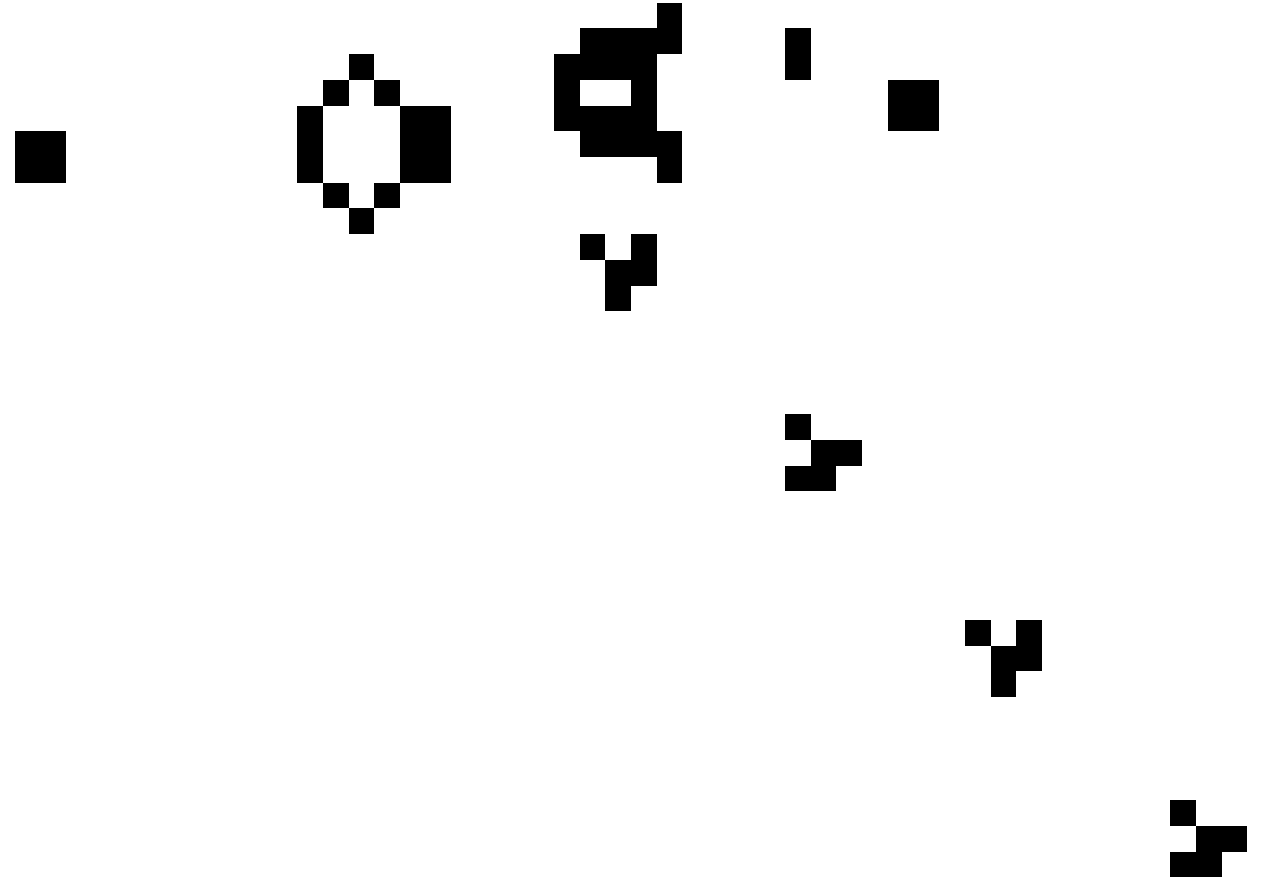
ON $\rightarrow$ OFF:

- if < 2 ON neighbours
- if > 3 ON neighbours

OFF $\rightarrow$ ON:

- if 3 ON neighbours

(otherwise cell does not change)



Turing patterning

MODEL

state $s_j(t)$

1 : ALIVE

0 : DEAD

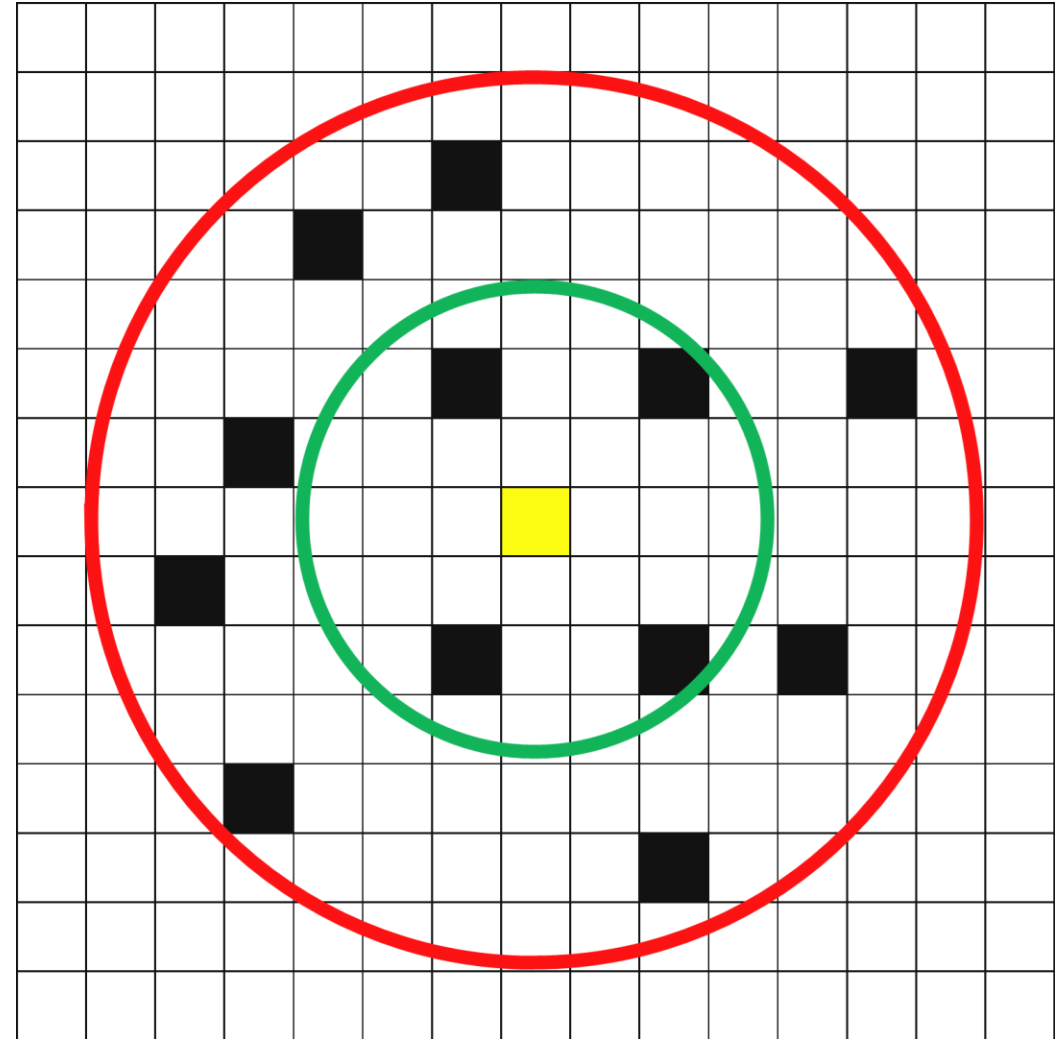
Update rule

Per cell, compute

$$S := \sum_{\mathcal{N}_a} w_a \frac{s_j}{|\mathcal{N}_a|} - \sum_{\mathcal{N}_i} w_i \frac{s_j}{|\mathcal{N}_i|}$$

If $S > 0$: cell becomes ALIVE

If $S \leq 0$: cell becomes DEAD



scale-dependent feedback

Turing patterning

MODEL

state $s_j(t)$

1 : ALIVE

0 : DEAD

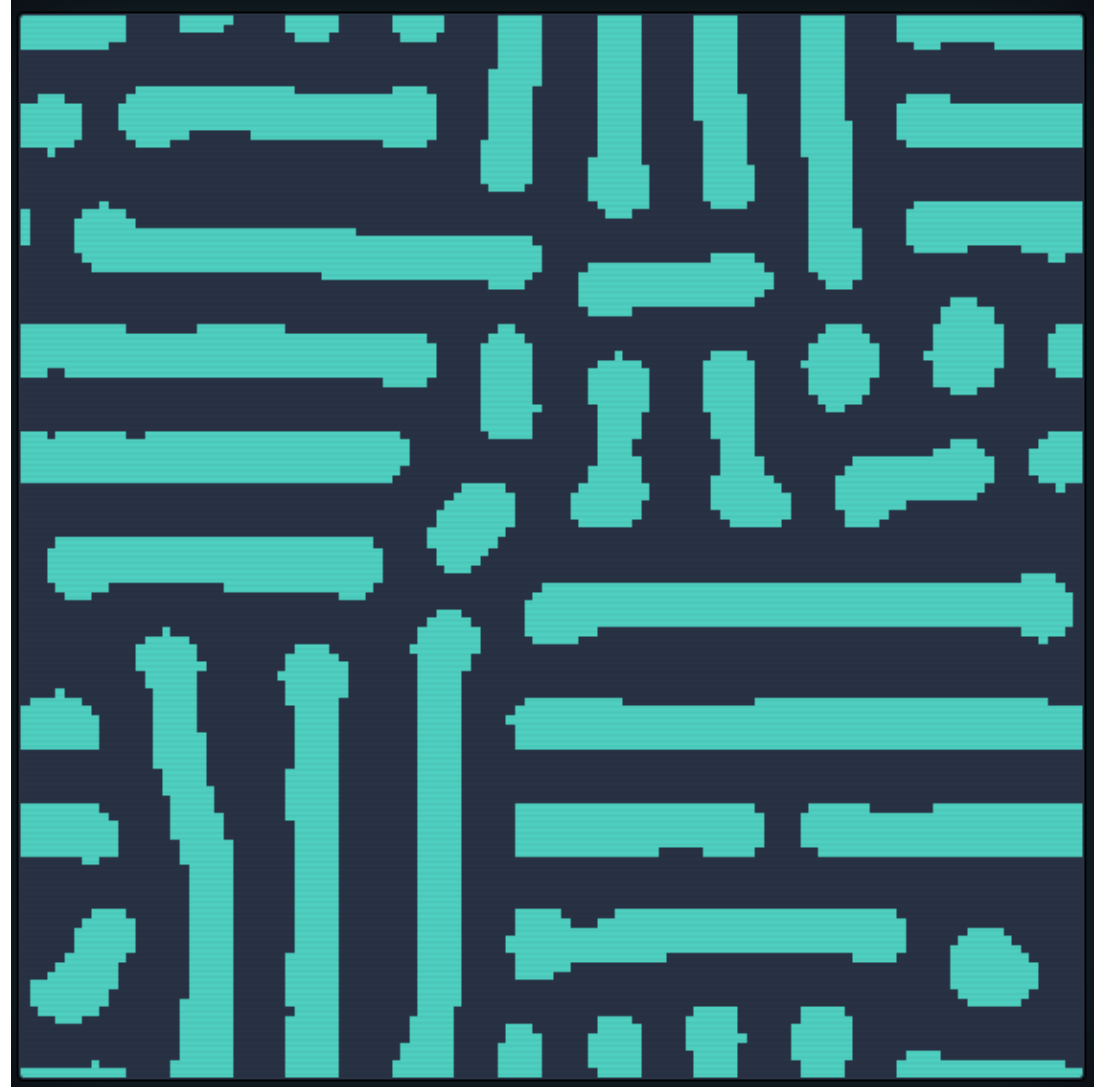
Update rule

Per cell, compute

$$S := \sum_{\mathcal{N}_a} w_a \frac{s_j}{|\mathcal{N}_a|} - \sum_{\mathcal{N}_i} w_i \frac{s_j}{|\mathcal{N}_i|}$$

If $S > 0$: cell becomes ALIVE

If $S \leq 0$: cell becomes DEAD



Parameters: $r_a = 3, w_a = 1,15, r_i = 6, w_i = 1.31$

Implemented at bastiaansen.github.io/CCSS_CA.html

Island model (only local interaction)



MODEL

state $s_j(t)$

0 : no vegetation

1 : vegetation

Windblown

$$\mathbb{P}(0 \rightarrow 1) = \epsilon$$

Reproduction

$$\mathbb{P}(0 \rightarrow 1) = \alpha \rho^2$$

with ρ density around site

Mortality

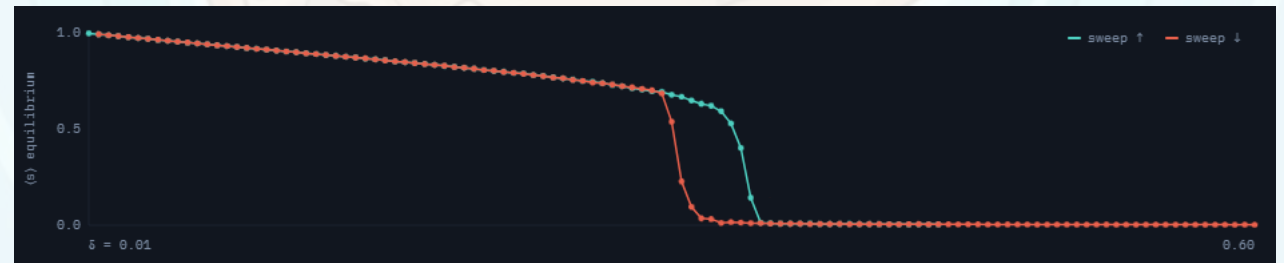
$$\mathbb{P}(1 \rightarrow 0) = \delta$$

Global interaction



Parameters: $\epsilon = 0.001, \alpha = 1.50, p = 1$ (use $\delta = 0.33$ to see differences)

Only local interaction





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A close-up photograph of a large number of ants, likely red ants, crawling along a green leaf. The ants are densely packed in some areas, particularly near the edge of the leaf, and are moving in various directions. The background is a soft, out-of-focus green.

Part VI

Other modelling frameworks

Agent-based / individual-based model (ABM/IBM)

MODEL (from agents.jl)

Each agent has location and velocity

Cohesion

- Move towards others

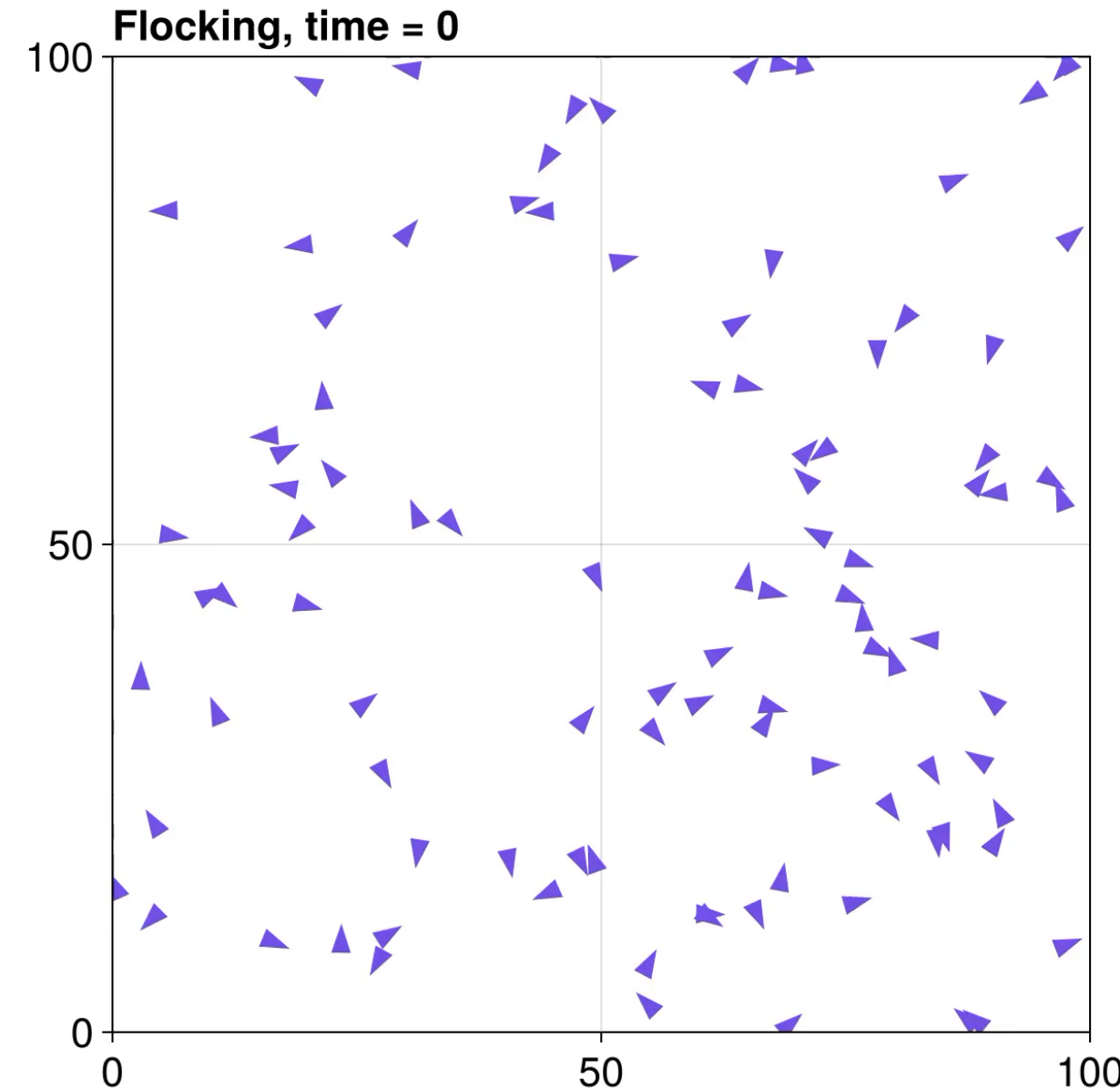
Separation

- Keep minimal distance to others

Alignment

- Match velocity of others

See <https://juliadynamics.github.io/Agents.jl/stable/examples/flock/>



Partial differential equations

MODEL

Gray-Scott model

$$\frac{\partial U}{\partial t} = D_u \nabla^2 U - UV^2 + F(1 - U)$$

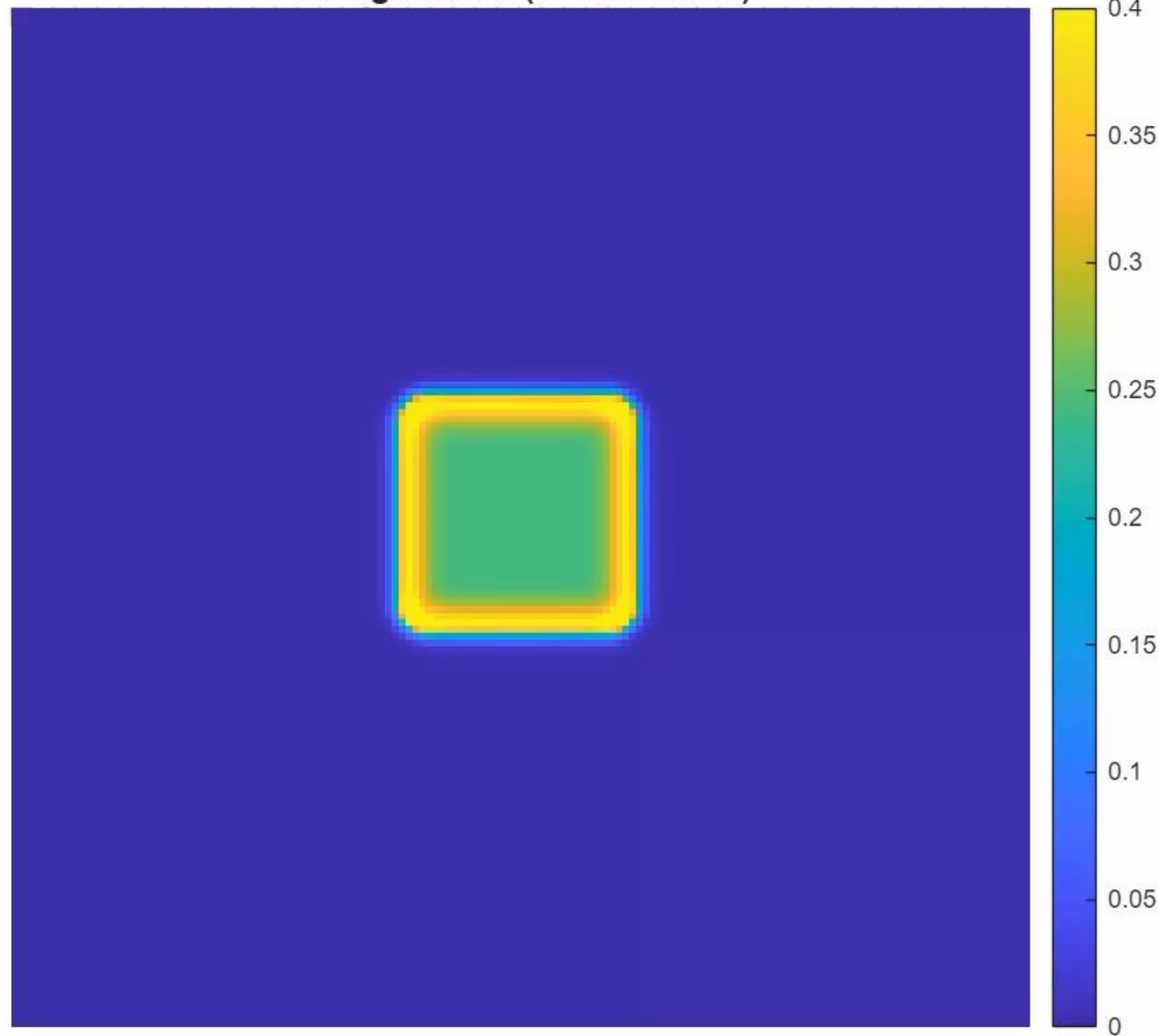
$$\frac{\partial V}{\partial t} = D_v \nabla^2 V + UV^2 - (F + k)V$$

$$D_u = 0.16, D_v = 0.08,$$

$$F = 0.04, k = 0.06$$

Can be derived as spatially-explicit
mean-field approximation

Turing Pattern (t = 40 / 10000)





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Part VII
Ok – so what?

Micro and macro

- Which micro rules lead to observed macro behaviour?

Examples:

- How does flocking work?
- Why are there spatial patterns?

- What macro behaviour emerges from observed micro dynamics?

Examples:

- What effects do pheromone deposits have in an ant colony?
- How does algorithmic trading influence financial markets?

How to study

- Numerical simulations
 - Cellular Automata (CA)
 - Agent-Based Models (ABMs / IBMs)
 - Ordinary or Partial Differential Equations (ODEs & PDEs)
- Analytic results only sometimes available
- Often: system-specific (data) analysis is required



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Feedbacks & Emergence

Summary

Feedbacks

negative/stabilising & positive/destabilising

Mean-field approximations

bifurcations/transitions of macro dynamics

Spatial structures

deviations from mean-field approximation
emergence of spatial structures